

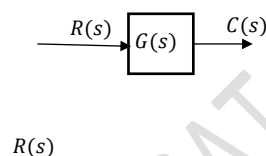
## Signal Flow Graph

Signal flow graphs are alternative to the block diagrams. Block diagrams consist of blocks (systems represented by transfer functions), signals, summing points, and pick off points. But, signal flow graph consists of **branches** and **nodes**. **Branches represent the systems (transfer functions)** and **nodes represent the signals**.

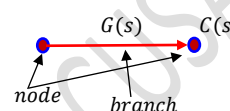
As we are dealing with linear systems, the input-output relation for each block will be linear.

$$C(s) = G(s)R(s)$$

For the complete system, we will get a number of such linear equations.



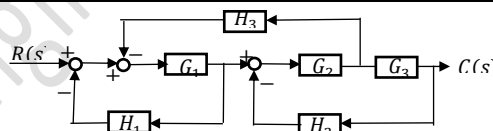
Signal flow graphs are graphical representation of linear equations. The above equation is represented as a signal flow graph as shown.



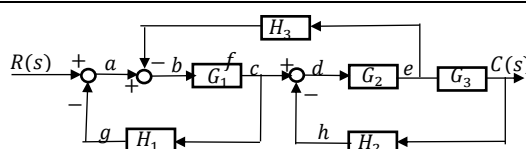
The input and output points or junctions are called **nodes**. All branches leaving a node will pass on the nodal signal signal to the output node all of each branch unidirectionally. The summation of all variables entering a node is equal to the node variable. A **path** is a branch or a continuous sequence of branches from one node to another. A **loop** is a closed path that originates from a node and terminates on the same node, with no node being met twice along the path. Two loops are said to be **nontouching** if they do not have a common node.

### Example 1

Consider the block diagram representation of a control system shown. Draw the signal flow graph.

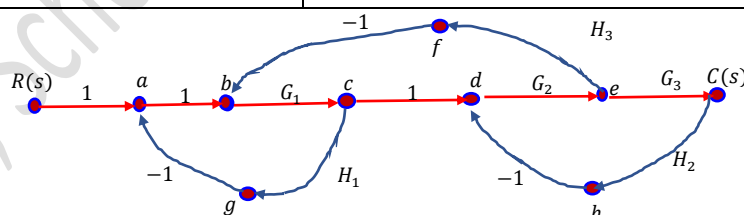


Let  $a, b, c, \dots$  represent the output signals from each functional block. Let us write the linear input-output relations for each of these blocks.



$$\begin{aligned} a &= R(s) - g \\ b &= a - f \\ c &= b \cdot G_1 \\ d &= c - h \\ e &= d \cdot G_2 \\ C(s) &= e \cdot G_3 \end{aligned}$$

$$\begin{aligned} g &= c \cdot H_1 \\ f &= e \cdot H_3 \\ h &= C(s) \cdot H_2 \end{aligned}$$



In signal flow graph, these signals are represented by nodes. Branches represent the transfer functions of the blocks. The values written on the branches like  $(1, -1, G_1, H_1 \text{ etc})$  are known as **branch gains**. The signal flow graph has **one path** from the input node to the output node.

$$P^1 \Rightarrow R(s) - a - b - c - d - e - C(s)$$

The path gain  $P^1$  is the product of all branch gains in the path.  $P^1 = 1 \times 1 \times G_1 \times 1 \times G_2 \times G_3 = G_1 G_2 G_3$

There are three loops:

$$\begin{aligned} L_1 &\Rightarrow a - b - c - g - a \\ L_2 &\Rightarrow b - c - d - e - f - b \\ L_3 &\Rightarrow d - e - C(s) - h - d \end{aligned}$$

The loop gain is the product of all branch gains in the loop.

$$\begin{aligned} L_1 &= 1 \times G_1 \times H_1 \times -1 = -G_1 H_1 \\ L_2 &= G_1 \times 1 \times G_2 \times H_3 \times -1 = -G_1 G_2 H_3 \\ L_3 &= G_2 \times G_3 \times H_2 \times -1 = -G_2 G_3 H_2 \end{aligned}$$

Since there are common nodes, loops  $L_1$  and  $L_2$  are touching each other like loops  $L_2$  and  $L_3$ . The loops  $L_1$  and  $L_3$  are nontouching.

The signal flow graph is a pictorial representation of the linear algebraic expressions showing the interdependencies of variables.

### Mason's signal flow gain formula

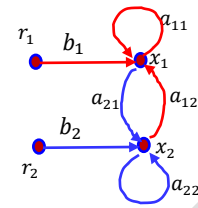
Consider the following set of simultaneous algebraic expressions

$$b_1 r_1 + a_{11} x_1 + a_{12} x_2 = x_1$$

$$b_2 r_2 + a_{21} x_1 + a_{22} x_2 = x_2$$

The input variables are  $r_1$  and  $r_2$ , and the output variables are  $x_1$  and  $x_2$ .

The signal flow graph for the above equations is shown in the figure.



There are two paths each from the input side to each output

$$P_{11}^1 \text{ (path 1 from input 1 to output 1)} \Rightarrow r_1 \rightarrow x_1$$

$$\text{Path gain, } P_{11}^1 = b_1$$

$$P_{21}^2 \text{ (path 2 from input 2 to output 1)} \Rightarrow r_2 \rightarrow x_2 \rightarrow x_1$$

$$\text{Path gain, } P_{21}^2 = b_2 \cdot a_{12} = b_2 a_{12}$$

$$P_{12}^1 \text{ (path 1 from input 1 to output 2)} \Rightarrow r_1 \rightarrow x_1 \rightarrow x_2$$

$$\text{Path gain, } P_{12}^1 = 1 \cdot a_{21} = b_1 a_{21}$$

$$P_{22}^2 \text{ (path 2 from input 2 to output 2)} \Rightarrow r_2 \rightarrow x_2$$

$$\text{Path gain, } P_{22}^2 = b_2$$

There are three loops in the signal flow graph with loop gains,  $L_1 = a_{11}$ ,  $L_2 = a_{22}$ , and  $L_3 = a_{12} a_{21}$ . Of these, loops  $L_1$  and  $L_2$  are not touching each other.

The equations are re-written as

$$b_1 r_1 = (1 - a_{11}) x_1 + (-a_{12}) x_2$$

$$b_2 r_2 = (-a_{21}) x_1 + (1 - a_{22}) x_2$$

$$\begin{bmatrix} (1 - a_{11}) & (-a_{12}) \\ (-a_{21}) & (1 - a_{22}) \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} = \begin{Bmatrix} b_1 r_1 \\ b_2 r_2 \end{Bmatrix}$$

$$x_1 = \frac{\begin{vmatrix} b_1 r_1 & -a_{12} \\ b_2 r_2 & (1 - a_{22}) \end{vmatrix}}{\begin{vmatrix} (1 - a_{11}) & -a_{12} \\ (-a_{21}) & (1 - a_{22}) \end{vmatrix}} = \frac{b_1(1 - a_{22})r_1 + b_2 a_{12} r_2}{(1 - a_{11})(1 - a_{22}) - (-a_{12} a_{21})}$$

$$x_2 = \frac{\begin{vmatrix} (1 - a_{11}) & b_1 r_1 \\ (-a_{21}) & b_2 r_2 \end{vmatrix}}{\begin{vmatrix} (1 - a_{11}) & -a_{12} \\ (-a_{21}) & (1 - a_{22}) \end{vmatrix}} = \frac{(1 - a_{11})b_2 r_2 - (-a_{21})b_1 r_1}{(1 - a_{11})(1 - a_{22}) - (-a_{12} a_{21})}$$

$$x_1 = \frac{b_1(1 - a_{22})r_1 + b_2 a_{12} r_2}{1 - (a_{11} + a_{22} + a_{12} a_{21}) + a_{11} a_{22}} = \frac{b_1(1 - a_{22})}{\Delta} r_1 + \frac{(b_2 a_{12}) \cdot 1}{\Delta} r_2$$

$$x_2 = \frac{b_1(a_{21})r_1 + b_2(1 - a_{11})r_2}{1 - (a_{11} + a_{22} + a_{12} a_{21}) + a_{11} a_{22}} = \frac{(b_1 a_{21}) \cdot 1}{\Delta} r_1 + \frac{b_2(1 - a_{11})}{\Delta} r_2$$

The denominator  $\Delta$  is the determinant given by

$$\Delta = 1 - (\text{sum of all loop gains}) + (\text{product of all combinations of two nontouching loops})$$

$$\Delta = 1 - \sum_{i=1}^N L_i + \sum_{\substack{i,j \\ \text{nontouching}}} L_i L_j - \sum_{\substack{i,j,k \\ \text{nontouching}}} L_i L_j L_k + \dots$$

The numerator for  $x_1$  with the input  $r_1$  is  $b_1$  times  $(1 - a_{22})$  which is the product of the gain of path 1 from input 1 to output 1 ( $P_{11}^1$ ) and the value of  $\Delta$  after eliminating all the loops touching that path ( $\Delta_{11}^1$ ). The numerator for  $x_1$  with the input  $r_2$  is  $(b_2 a_{12})$  times  $(1)$  which is the product of the gain of path 2 from input 2 to output 1 ( $P_{21}^2$ ) and the value of  $\Delta$  after eliminating all the loops touching that path ( $\Delta_{21}^2$ ).

Similarly, the numerator for  $x_2$  with the input  $r_1$  is  $(b_1 a_{21})$  times  $(1)$  which is the product of the gain of path 1 from input 1 to output 2 ( $P_{12}^1$ ) and the value of  $\Delta$  after eliminating all the loops touching that path ( $\Delta_{12}^1$ ). The numerator for  $x_2$  with the input  $r_2$  is  $b_2$  times  $(1 - a_{11})$  which is the product of the gain of path 2 from input 2 to output 2 ( $P_{22}^2$ ) and the value of  $\Delta$  after eliminating all the loops touching that path ( $\Delta_{22}^2$ ).

$$x_1 = \frac{P_{11}^1 \cdot \Delta_{11}^1 r_1 + P_{21}^2 \cdot \Delta_{21}^2 r_2}{\Delta} \quad \ll \gg \quad x_2 = \frac{P_{12}^1 \cdot \Delta_{12}^1 r_1 + P_{22}^2 \cdot \Delta_{22}^2 r_2}{\Delta}$$

If we have a single input  $R(s)$ -single output  $C(s)$  system with different paths from input to output

$$C(s) = \frac{P^1 \Delta^1 R(s) + P^2 \Delta^2 R(s) + \dots}{\Delta} = \left( \frac{P^1 \Delta^1 + P^2 \Delta^2 + \dots}{\Delta} \right) R(s)$$

The transfer function of the system is given by,

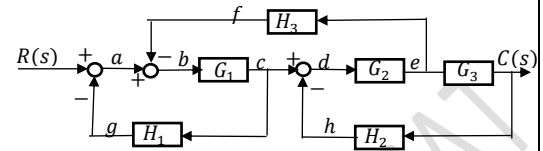
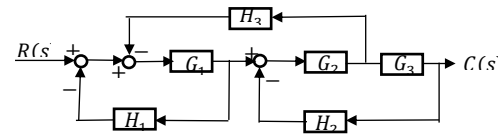
$$G(s) = \frac{C(s)}{R(s)} = \left( \frac{P^1 \Delta^1 + P^2 \Delta^2 + \dots}{\Delta} \right) = \frac{\sum_k P^k \Delta^k}{\Delta}$$

This is known as **Mason's Signal flow gain formula**.

### Example 2

Consider the block diagram representation of a control system shown. Draw the signal flow graph and obtain the overall transfer function of the system.

Let  $a, b, c, \dots$  represent the output signals from each functional block. Let us write the linear input-output relations for each of these blocks.



$$a = R(s) - g$$

$$g = c \cdot H_1$$

$$b = a - f$$

$$f = e \cdot H_3$$

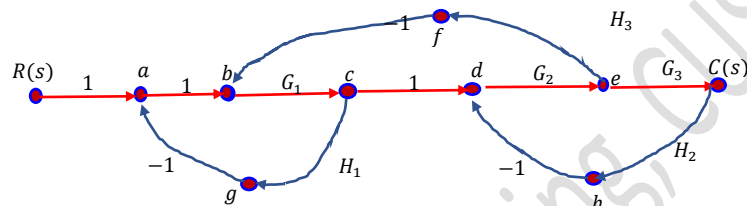
$$c = b \cdot G_1$$

$$d = c - h$$

$$h = C(s) \cdot H_2$$

$$e = d \cdot G_2$$

$$C(s) = e \cdot G_3$$



The signal flow graph has **one path** from the input node to the output node.

$$P^1 \Rightarrow R(s) - a - b - c - d - e - C(s)$$

The path gain  $P^1$  is the product of all branch gains in the path.  $P^1 = 1 \times 1 \times G_1 \times 1 \times G_2 \times G_3 = G_1 G_2 G_3$

There are three loops:

$$L_1 \Rightarrow a - b - c - g - a$$

$$L_2 \Rightarrow b - c - d - e - f - b$$

$$L_3 \Rightarrow d - e - C(s) - h - d$$

The loop gain is the product of all branch gains in the loop.

$$L_1 = 1 \times G_1 \times H_1 \times -1 = -G_1 H_1$$

$$L_2 = G_1 \times 1 \times G_2 \times H_3 \times -1 = -G_1 G_2 H_3$$

$$L_3 = G_2 \times G_3 \times H_2 \times -1 = -G_2 G_3 H_2$$

Since there are common nodes, loops  $L_1$  and  $L_2$  are touching each other like loops  $L_2$  and  $L_3$ . The loops  $L_1$  and  $L_3$  are **nontouching**.

$$\Delta = 1 - \sum_{i=1}^N L_i + \sum_{\substack{i,j \\ \text{nontouching}}} L_i L_j - \sum_{\substack{i,j,k \\ \text{nontouching}}} L_i L_j L_k + \dots$$

$$\Delta = 1 - (L_1 + L_2 + L_3) + (L_1 L_3) = 1 + G_1 H_1 + G_1 G_2 H_3 + G_2 G_3 H_2 + G_1 G_2 G_3 H_1 H_2$$

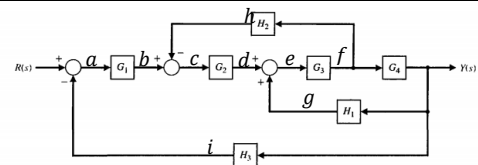
$\Delta^1$  is the value of  $\Delta$  after eliminating all the loops touching the path 1 (All loops are touching  $P^1$ ).

$$\Delta^1 = 1$$

$$G(s) = \frac{C(s)}{R(s)} = \frac{P^1 \Delta^1}{\Delta} = \frac{G_1 G_2 G_3}{1 + G_1 H_1 + G_1 G_2 H_3 + G_2 G_3 H_2 + G_1 G_2 G_3 H_1 H_2}$$

**Example 3**

Draw the signal flow graph and find the overall transfer function of the system.



$$a = R(s) - i$$

$$i = Y(s) \cdot H_3$$

$$b = a G_1$$

$$h = f \cdot H_2$$

$$c = b - h$$

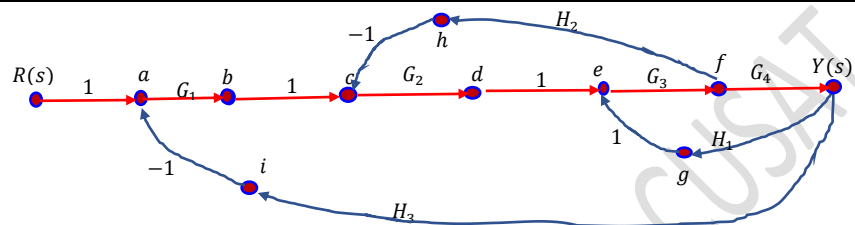
$$d = c \cdot G_2$$

$$g = Y(s) \cdot H_1$$

$$e = d + g$$

$$f = e G_3$$

$$Y(s) = f \cdot G_4$$



The signal flow graph has **one path** from the input node to the output node.

$$P^1 \Rightarrow R(s) - a - b - c - d - e - f - Y(s)$$

The path gain  $P^1$  is the product of all branch gains in the path.  $P^1 = 1 \times G_1 \times 1 \times G_2 \times 1 \times G_3 \times G_4 = G_1 G_2 G_3 G_4$

There are three loops:

$$L_1 \Rightarrow a - b - c - d - e - f - Y(s) - i - a$$

The loop gain is the product of all branch gains in the loop.

$$L_2 \Rightarrow c - d - e - f - h - c$$

$$L_1 = G_1 \times 1 \times G_2 \times 1 \times G_3 \times G_4 \times H_3 \times -1 = -G_1 G_2 G_3 G_4 H_3$$

$$L_3 \Rightarrow e - f - Y(s) - g - e$$

$$L_2 = G_2 \times 1 \times G_3 \times H_2 \times -1 = -G_2 G_3 H_2$$

$$L_3 = G_3 \times G_4 \times H_1 \times 1 = +G_3 G_4 H_1$$

Since there are common nodes, loops  $L_1$  and  $L_2$  are touching each other like loops  $L_2$  and  $L_3$ , and loops  $L_1$  and  $L_3$ . There are **no nontouching loops**.

$$\Delta = 1 - \sum_{i=1}^N L_i + \sum_{\substack{i,j \\ \text{nontouching}}} L_i L_j - \sum_{\substack{i,j,k \\ \text{nontouching}}} L_i L_j L_k + \dots$$

$$\Delta = 1 - (L_1 + L_2 + L_3) = 1 + G_1 G_2 G_3 G_4 H_3 + G_2 G_3 H_2 - G_3 G_4 H_1$$

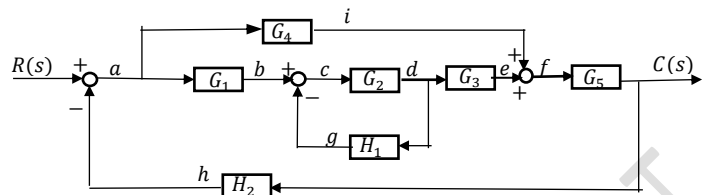
$\Delta^1$  is the value of  $\Delta$  after eliminating all the loops touching the path 1 (All loops are touching  $P^1$ ).

$$\Delta^1 = 1$$

$$G(s) = \frac{C(s)}{R(s)} = \frac{P^1 \Delta^1}{\Delta} = \frac{G_1 G_2 G_3 G_4}{1 - G_3 G_4 H_1 + G_2 G_3 H_2 + G_1 G_2 G_3 G_4 H_3}$$

**Example 4**

Draw the signal flow graph and find the overall transfer function of the system.



$$a = R(s) - h$$

$$h = C(s) \cdot H_2$$

$$b = aG_1$$

$$g = d \cdot H_1$$

$$c = b - g$$

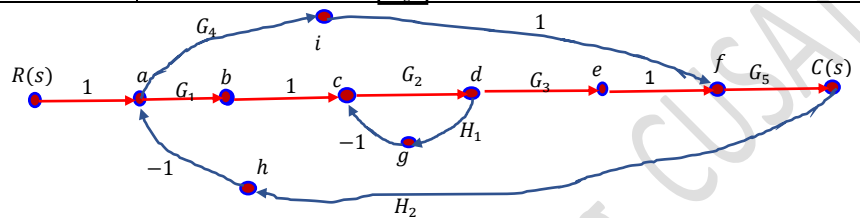
$$d = c \cdot G_2$$

$$e = dG_3$$

$$f = e + i$$

$$i = aG_4$$

$$C(s) = f \cdot G_5$$



The signal flow graph has **two paths** from the input node to the output node.

$$P^1 \Rightarrow R(s) - a - b - c - d - e - f - C(s)$$

$$\text{Path gains: } P^1 = 1 \times G_1 \times 1 \times G_2 \times 1 \times G_3 \times G_5 = G_1 G_2 G_3 G_5$$

$$P^2 \Rightarrow R(s) - a - i - f - C(s)$$

$$P^2 = 1 \times G_4 \times 1 \times G_5 = G_4 G_5$$

There are three loops:

$$L_1 \Rightarrow c - d - g - c$$

$$\text{Loop gains: } L_1 = G_2 \times H_1 \times -1 = -G_2 H_1$$

$$L_2 \Rightarrow a - b - c - d - e - f - C(s) - h - a$$

$$L_2 = G_1 \times 1 \times G_2 \times 1 \times G_3 \times G_5 \times H_2 \times -1 = -G_1 G_2 G_3 G_5 H_2$$

$$L_3 \Rightarrow a - i - f - C(s) - h - a$$

$$L_3 = G_4 \times 1 \times G_5 \times H_2 \times -1 = -G_4 G_5 H_2$$

Since there are common nodes, loops  $L_1$  and  $L_2$  are touching each other like loops  $L_2$  and  $L_3$ , and loops  $L_1$  and  $L_3$  are **nontouching loops**.

$$\Delta = 1 - \sum_{i=1}^N L_i + \sum_{\substack{i,j \\ \text{nontouching}}} L_i L_j - \sum_{\substack{i,j,k \\ \text{nontouching}}} L_i L_j L_k + \dots$$

$$\Delta = 1 - (L_1 + L_2 + L_3) + (L_1 L_3) = 1 + G_2 H_1 + G_1 G_2 G_3 G_5 H_2 + G_4 G_5 H_2 + G_2 G_4 G_5 H_1 H_2$$

$\Delta^1$  is the value of  $\Delta$  after eliminating all the loops touching the path 1 (All loops are touching  $P^1$ ).

$$\Delta^1 = 1$$

$\Delta^2$  is the value of  $\Delta$  after eliminating all the loops touching the path 2 (loops  $L_2$  and  $L_3$  are touching  $P^2$ ).

$$\Delta^2 = 1 - L_1 = 1 + G_2 H_1$$

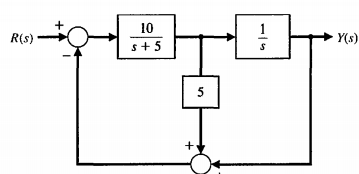
$$G(s) = \frac{C(s)}{R(s)} = \frac{P^1 \Delta^1 + P^2 \Delta^2}{\Delta} = \frac{G_1 G_2 G_3 G_5 + G_4 G_5 (1 + G_2 H_1)}{1 + G_2 H_1 + G_1 G_2 G_3 G_5 H_2 + G_4 G_5 H_2 + G_2 G_4 G_5 H_1 H_2}$$

$$G(s) = \frac{G_4 G_5 + G_1 G_2 G_3 G_5 + G_2 G_4 G_5 H_1}{1 + G_2 H_1 + G_4 G_5 H_2 + G_1 G_2 G_3 G_5 H_2 + G_2 G_4 G_5 H_1 H_2}$$

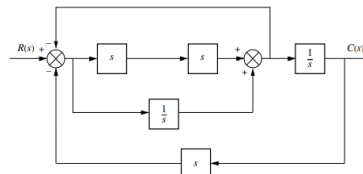
### Exercise

Draw the signal flow graph and determine the closed loop transfer function  $T(s) = \frac{Y(s)}{R(s)}$

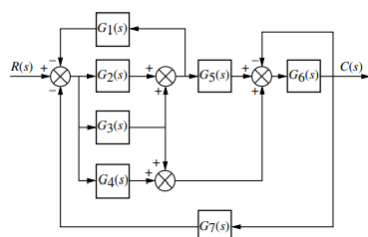
1.



2.



3.



4.

